X.Link-GW-ZB485

User manual

IIOT Device Gateway, Ethernet 10/100Mbps, Zigbee Mesh 2.4GHz, 4 Channel RS485

Overview

The IIOT Gateway is a compact and versatile industrial data acquisition and communication device designed for smart factory and automation systems. It operates with a wide input voltage range from 8 to 28 VDC, ensuring stable performance in various industrial environments.

The device features four independent RS485 channels, each configurable as Modbus RTU Master or Slave, enabling seamless integration with multiple PLCs, sensors, and controllers. It also includes a 2.4 GHz Zigbee Mesh interface for wireless sensor networking and extended coverage in distributed systems.

For high-speed wired communication, the gateway provides a 100 Mbps Ethernet port, supporting industrial protocols such as Modbus TCP, MQTT, HTTP, and FTP. These communication options allow flexible data transmission to SCADA systems, cloud platforms, or local servers, making it ideal for applications in industrial automation, energy management, and remote monitoring.

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1 Features

- High-performance MCUs based on Arm® Cortex® - M33 core with 2 Mbytes of Flash memory and 640 Kbytes of SRAM
- 8-Mbytes Quad-SPI (Winbond) Flash Memory
- Support up to 64-Gbytes SD Card
- Real Time Operating System
- 1 - Zigbee® mesh 2.4 GHz
- 3 - RS485 channel speeds up to 115200 bps
- 2 - CAN channel speeds up to 1 Mbps
- 1 - Ethernet speeds up to 100 Mbps
- 1 - USB type C configuration
- Firmware Over-the-Air (FOTA)
- Led indicator. 1-Led power, 1-Led status, 1-Led link
- Support MODBUS RTU master / slave
- Support MODBUS TCP
- Support MQTT/MQTTS HTTP/HTTPS FTP/FTPS

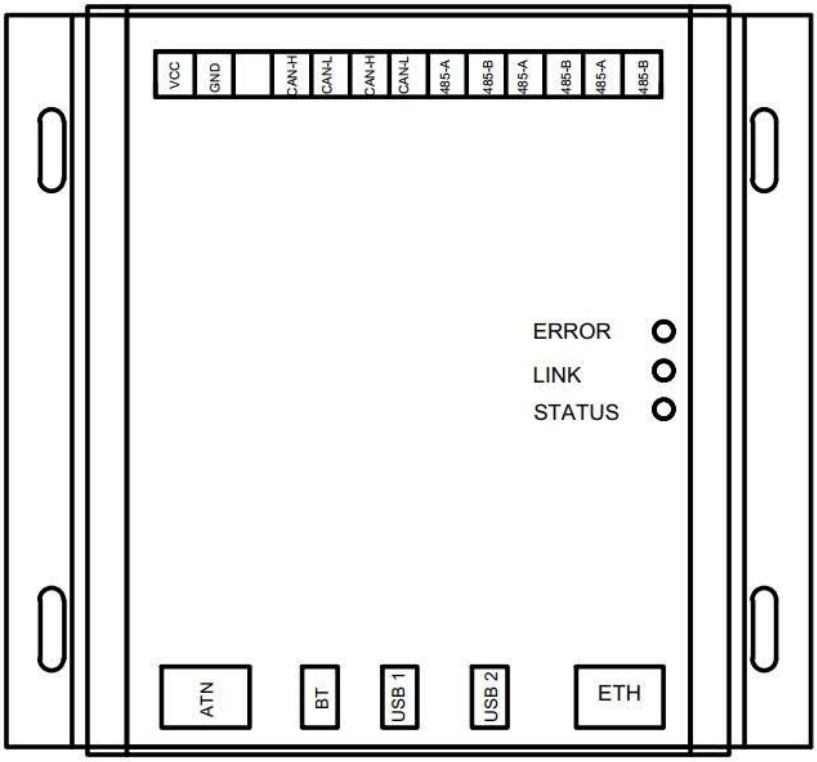
2 Specifications

		MIN	MAX	UNIT
Voltage range	Power voltage	8	28	V
	Isolated digital input voltage	8	28	V
	Isolated digital output voltage	8	28	V
Current range				
	Isolated digital input current	0.1	2	A
	Isolated digital output current	0.1	2	A
MTBF			250000	h
Temperature		-40	85	°C
Humidity		10	80	%

3 Applications

- Remote Machine control and monitoring
- Remote Counter
- Remote Tower lamp control
- Industrial Automation
- Smart monitoring

4 Pin Configuration and Functions



5 Modbus Register

Address	Size	Type	R/W	Units	Descriptions
10000	1	Coil Register	R/W	bit	Output 0
10001	1	Coil Register	R/W	bit	Output 1
10002	1	Coil Register	R/W	bit	Output 2
10003	1	Coil Register	R/W	bit	Output 3
10004	1	Coil Register	R/W	bit	Output 4
10005	1	Coil Register	R/W	bit	Output 5
10006	1	Coil Register	R/W	bit	Output 6
10007	1	Coil Register	R/W	bit	Output 7
10008	1	Coil Register	R/W	bit	Reset counter 0
10009	1	Coil Register	R/W	bit	Reset counter 1
10010	1	Coil Register	R/W	bit	Reset counter 2
10011	1	Coil Register	R/W	bit	Reset counter 3
10012	1	Coil Register	R/W	bit	Reset counter 4
10013	1	Coil Register	R/W	bit	Reset counter 5
10014	1	Coil Register	R/W	bit	Reset counter 6
10015	1	Coil Register	R/W	bit	Reset counter 7
20000	1	Discrete Register	R	bit	Input 0
20001	1	Discrete	R	bit	Input 1

		Register			
20002	1	Discrete Register	R	bit	Input 2
20003	1	Discrete Register	R	bit	Input 3
20004	1	Discrete Register	R	bit	Input 4
20005	1	Discrete Register	R	bit	Input 5
20006	1	Discrete Register	R	bit	Input 6
20007	1	Discrete Register	R	bit	Input 7
30000	16	Holding register	R/W	char	Device name
30016	16	Holding register	R/W	Integer	DHCP Enable
30017	16	Holding register	R/W	Integer	IP Address
30018	16	Holding register	R/W	Integer	Subnet mash
30019	16	Holding register	R/W	Integer	Default gateway
30020	16	Holding register	R/W	Integer	MAC [0,1]
30021	16	Holding register	R/W	Integer	MAC [2,3]
30022	16	Holding register	R/W	Integer	MAC [4,5]
30023	16	Holding register	R/W	char	MQTT HOST
30055	16	Holding register	R/W	Integer	MQTT PORT
30087	16	Holding register	R/W	Integer	User name
30119	16	Holding register	R/W	Integer	Pass word
30151	16	Holding register	R/W	Integer	Client ID
30152	16	Holding register	R/W	Integer	Publish Times
30153	16	Holding register	R/W	Integer	RS485-1 TCP PORT
30154	16	Holding register	R/W	Integer	RS485-2 TCP PORT
30155	16	Holding register	R/W	Integer	RS485-3 TCP PORT
30156	16	Holding register	R/W	Integer	RS485-1 Baud rate
30157	16	Holding register	R/W	Integer	RS485-2 Baud rate
30158	16	Holding register	R/W	Integer	RS485-3 Baud rate
30159	16	Holding register	R/W	Integer	CAN-1 Baud rate
30160	16	Holding register	R/W	Integer	CAN-2 Baud rate
30161	16	Holding register	R/W	Integer	Zigbee Point type
30162	16	Holding register	R/W	Integer	Zigbee PAN ID
30163	16	Holding register	R/W	Integer	Zigbee Channel
30164	16	Holding register	R/W	Integer	Zigbee User Address
30165	16	Holding register	R/W	Integer	Zigbee antenna selected
30166	16	Holding register	R/W	Integer	Zigbee security
30167	16	Holding register	R/W	Integer	Zigbee security code 0
30168	16	Holding register	R/W	Integer	Zigbee security code 1
30169	16	Holding register	R/W	Integer	Zigbee security code 2
30170	16	Holding register	R/W	Integer	Zigbee security code 3
30171	16	Holding register	R/W	Integer	Number of Zigbee IO device can connect
30172	16	Holding register	R/W	Integer	Start address Zigbee IO device

40000	8	Input register	R	string	Serial number
40016	16	Input register	R	Integer	Counter 0
40017	16	Input register	R	Integer	Counter 1
40018	16	Input register	R	Integer	Counter 2
40019	16	Input register	R	Integer	Counter 3
40020	16	Input register	R	Integer	Timer 0 counter
40021	16	Input register	R	Integer	Timer 1 counter
40022	16	Input register	R	Integer	Timer 2 counter
40023	16	Input register	R	Integer	Timer 3 counter

6 System requirements

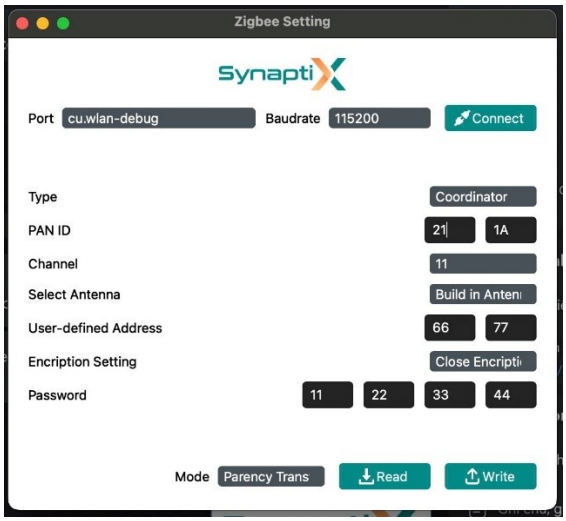
- Windows® OS (XP, 7, 8 and 10), Linux® or MacOS™
- USB Type-A to Type-C cable
- USB Driver

7 Configuration software

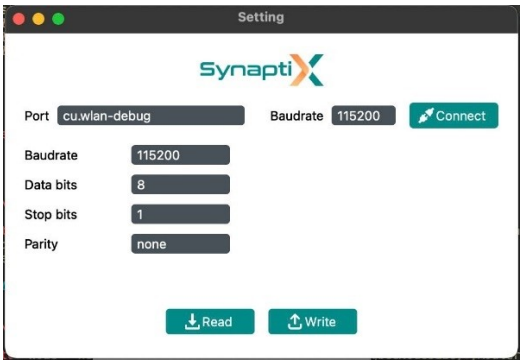
- Software : X.LinkConfig



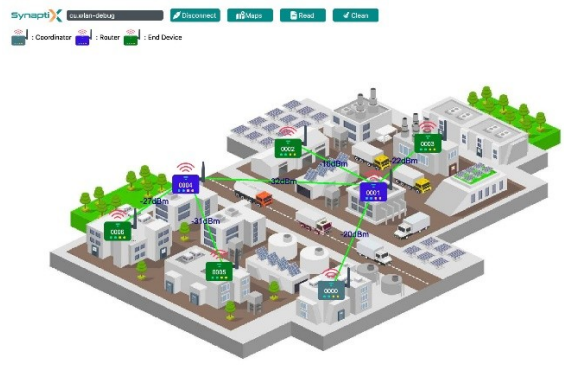
Main Selection Page



Zigbee Parameter Configuration Page



RS485 Configuration Page



Checking Signal Device Page (Only using with Gateway)

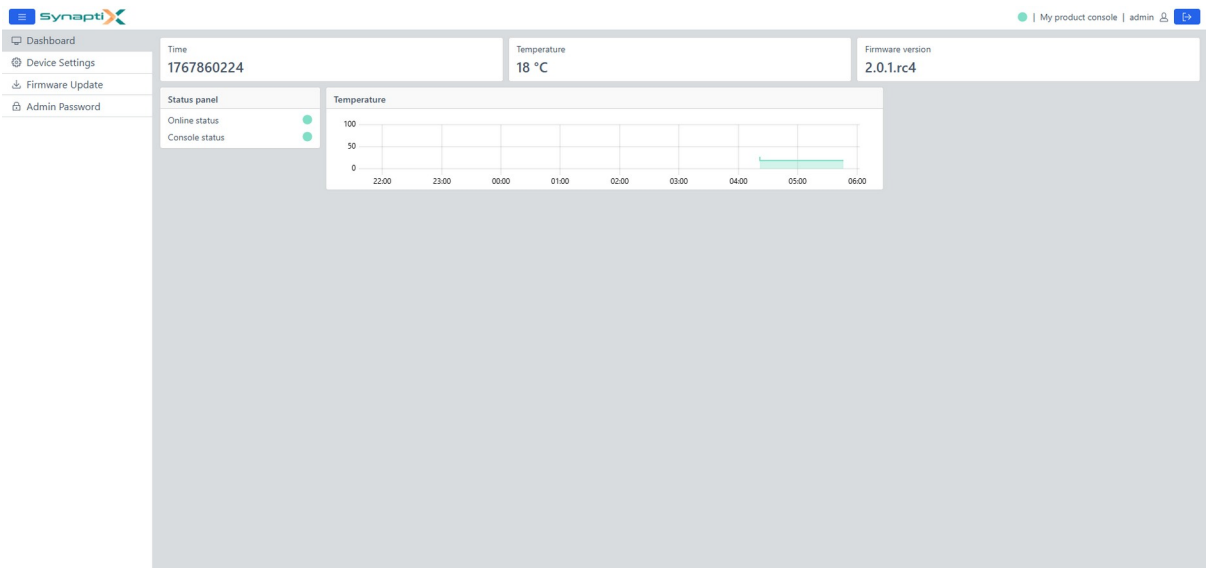
8 Website Configuration Page

1. Login Page

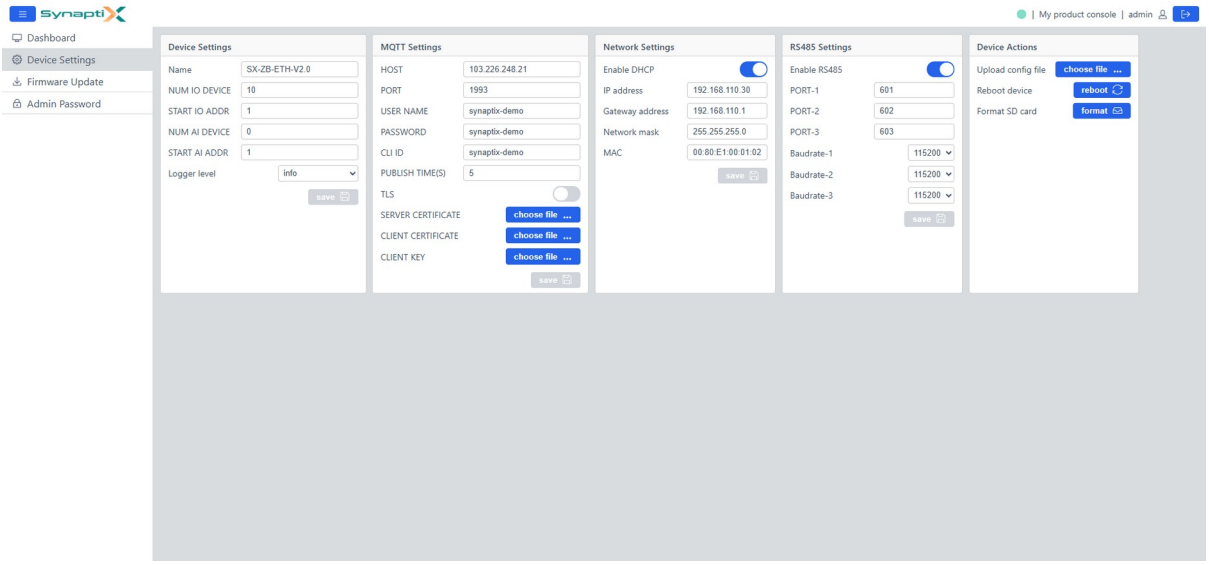


Default user : admin. Default password : admin

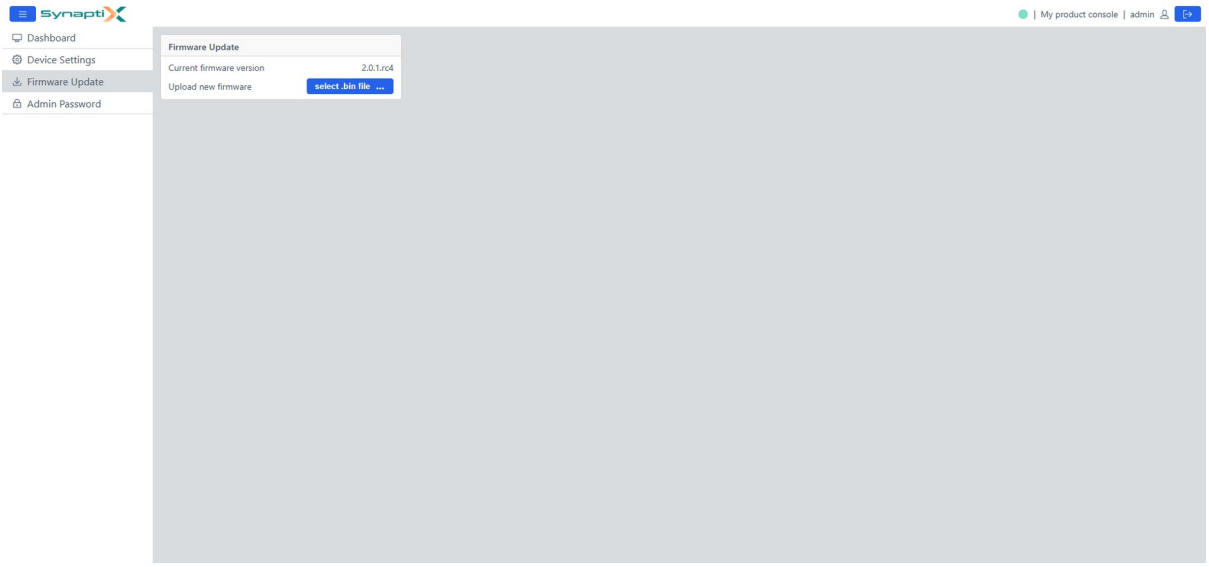
2. Dashboard



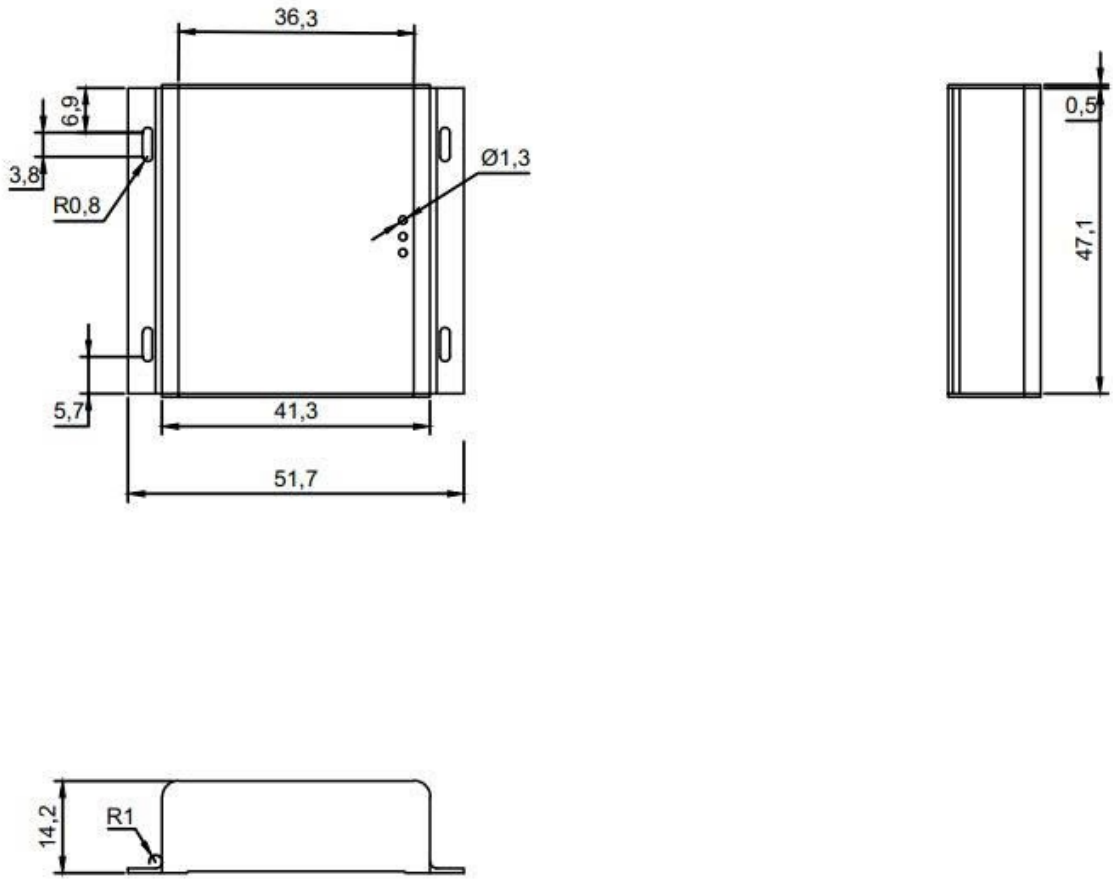
3. Device Settings



4. Firmware Update



9 Dimension



10 Ordering information

Part number	RS-485	RF type	Ethernet	USB
GW-ZB485	4 CH	Zigbee	1 CH	2 CH
GW-LR485	4 CH	LORA	1 CH	2 CH

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